



Summer Internship Project Report

Design and Implementation of a SPWM-Based Single-Phase Inverter

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Abstract

This report presents the complete design, implementation, and testing of a sinusoidal pulse width modulation (SPWM) based single-phase inverter. The system utilizes an STM32F103C8T6 microcontroller to generate complementary 18 kHz SPWM signals with programmable dead-time to safely drive an H-bridge MOSFET topology built around four IRF3205 N-channel power MOSFETs. An optocoupler-based gate drive architecture, employing TLP251 optocouplers and a bootstrapping technique using 100 μF capacitors, ensures efficient high-side switching while minimizing the number of isolated power supplies. A second-order LC low-pass filter (1 mH, 2.2 μF) is used to attenuate high-frequency switching harmonics. Experimental evaluation under a 2.2 Ω resistive load reveals performance limitations, including waveform distortion and reduced amplitude compared to the no-load condition. This project successfully demonstrates the complete workflow from schematic capture and PCB layout in EasyEDA to hardware assembly and oscilloscope-based debugging, providing insight into practical inverter design challenges, gate drive requirements, and filter optimization. The report concludes with recommendations for improving load regulation through closed-loop control and optimized filter design.

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Internship Organization and Project Relevance

This internship was conducted at the National Engineering and Scientific Commission (NESCOM), a premier research and development organization in Pakistan dedicated to advancing indigenous technological capabilities in defense, aerospace, and strategic sectors. NESCOM's mission encompasses the development of sophisticated electronic systems, power electronics, and control systems for various national applications, including radar systems, unmanned systems, communication networks, and renewable energy solutions. The organization places significant emphasis on in-house design and development of power conversion systems to ensure reliability, security, and performance in critical defense and civilian infrastructure. The project, "Design and Implementation of a SPWM-Based Single-Phase Inverter," aligns with NESCOM's research interests in power electronics and embedded systems. The ability to efficiently convert DC power to AC power with minimal distortion is fundamental to numerous applications, including satellite power systems, on-board vehicular power supplies, radar transmitter power stages, and emergency backup systems for strategic installations. This project provided the opportunity to develop practical expertise in microcontroller-based SPWM signal generation, gate drive design using optocouplers and bootstrapping techniques, and LC filter optimization, all of which are directly applicable to real-world power electronics challenges.

Chapter 1

Introduction

1.1 Background

The growing reliance on renewable energy sources, uninterruptible power supplies (UPS), and portable electronics has intensified the demand for efficient and compact DC-AC power conversion systems. Pulse width modulation (PWM) inverters are at the heart of this technology, converting a direct current (DC) input into an alternating current (AC) output. Among various modulation strategies, sinusoidal pulse width modulation (SPWM) stands out due to its ability to generate a low-distortion, low-harmonic AC waveform by varying the pulse widths proportionally to a reference sine wave [3].

1.2 Problem Statement

The primary challenge in designing such inverters lies in ensuring safe switching transitions to prevent catastrophic "shoot-through" conditions while maintaining a clean sinusoidal output under varying load conditions. Additionally, the output voltage must remain stable regardless of load variations, requiring careful design of the power stage, gate drive circuitry, and output filtering.

1.3 Project Scope

This project aims to design, simulate, and physically implement a low-cost SPWM inverter. The heart of the controller is the STM32F103C8T6 microcontroller, which allows precise generation of complementary SPWM signals. The hardware incorporates TLP251 optocouplers for gate isolation and bootstrapping networks for high-side drive. The final stage includes an LC filter to smooth the output. This report details the design methodology, hardware implementation (including PCB design in EasyEDA), and experimental validation under resistive loading, critically analyzing the deviations observed between theoretical projections and real-world performance.

Chapter 2

Literature Review

2.1 Introduction to SPWM Inverters

Sinusoidal Pulse Width Modulation (SPWM) is a widely used technique for DC-AC power conversion. The basic principle involves comparing a high-frequency triangular carrier wave with a low-frequency sinusoidal reference wave. The intersection points determine the switching instants, producing a train of pulses whose width varies sinusoidally [3].

The output voltage of a single-phase SPWM inverter can be expressed as:

$$V_o(t) = m_a V_{dc} \sin(\omega t) + \sum_{n=1}^{\infty} V_n \sin(n\omega_c t \pm \omega t) \quad (2.1)$$

where m_a is the modulation index, V_{dc} is the DC bus voltage, ω is the fundamental frequency, and ω_c is the carrier frequency.

2.1.1 Modulation Index

The modulation index (m_a) is defined as the ratio of the amplitude of the reference sine wave to the amplitude of the carrier wave:

$$m_a = \frac{V_{ref}}{V_{carrier}} \quad (2.2)$$

For linear modulation ($m_a \leq 1$), the fundamental output voltage is directly proportional to the modulation index. Over-modulation ($m_a > 1$) increases the fundamental voltage but introduces additional harmonics.

2.1.2 Carrier Frequency

The carrier frequency (also called switching frequency) determines the number of pulses per cycle. Higher carrier frequencies produce better waveform quality (lower THD) but increase switching losses. Typical switching frequencies range from 5 kHz to 20 kHz for IGBT-based inverters and up to 100 kHz for MOSFET-based designs [3].

2.2 Dead-Time in SPWM Inverters

Dead-time is a critical requirement in inverter design to prevent shoot-through, where both switches in a half-bridge conduct simultaneously, shorting the DC bus. The dead-time is inserted between the turn-off of one switch and the turn-on of the complementary switch [5].

2.2.1 Effects of Dead-Time

While dead-time is essential for safety, it introduces several undesirable effects [5]:

1. **Voltage Error:** The output voltage deviates from the ideal sinusoidal waveform, introducing distortion.
2. **Amplitude Reduction:** The effective output voltage decreases, reducing the inverter's voltage utilization.
3. **Low-Order Harmonics:** Dead-time introduces odd-order harmonics, particularly the 3rd, 5th, and 7th harmonics.
4. **Zero-Crossing Distortion:** The most significant distortion occurs at the zero-crossing of the output current.

2.2.2 Dead-Time Compensation

Various compensation techniques have been proposed to mitigate dead-time effects [4]:

1. **Current Polarity-Based Compensation:** The compensation voltage is added or subtracted based on the direction of load current.
2. **Predictive Compensation:** The dead-time voltage error is predicted and pre-compensated in the modulation.
3. **Adaptive Compensation:** Real-time measurement of the dead-time voltage error and dynamic adjustment.

2.3 LC Filter Design

The output LC filter is essential for converting the high-frequency SPWM waveform into a smooth sinusoidal voltage. The filter attenuates the carrier frequency components while passing the fundamental frequency [3].

2.3.1 Filter Transfer Function

The transfer function of a second-order LC filter is:

$$H(s) = \frac{1}{1 + s^2 LC} \quad (2.3)$$

The cutoff frequency is given by:

$$f_c = \frac{1}{2\pi\sqrt{LC}} \quad (2.4)$$

2.3.2 Damping Considerations

The damping factor of the LC filter is determined by the load resistance:

$$\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}} \quad (2.5)$$

For critical damping, $\zeta = 1$. Under-damped filters ($\zeta < 1$) exhibit ringing, while over-damped filters ($\zeta > 1$) have slower response but better stability.

2.4 Gate Drive Design

Gate drive circuitry is critical for proper MOSFET operation. The key requirements include [3]:

1. **Sufficient Gate Voltage:** Typically 10 V to 15 V for MOSFETs.
2. **High Peak Current:** To charge/discharge the gate capacitance rapidly.
3. **Galvanic Isolation:** Between the control circuit and power stage.
4. **Bootstrapping:** For high-side MOSFET drive.

2.5 Total Harmonic Distortion (THD)

THD is a key performance metric for inverters, defined as:

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} V_n^2}}{V_1} \times 100\% \quad (2.6)$$

where V_1 is the fundamental voltage and V_n are the harmonic voltages. IEEE Standard 519 recommends THD limits for various applications, typically less than 5% for general-purpose inverters.

Chapter 3

System Architecture and Design Methodology

3.1 System Overview

The entire system is built upon a modular architecture:

- **Control Stage:** STM32F103C8T6 ARM Cortex-M3 microcontroller operating at 18 kHz switching frequency.
- **Isolation Stage:** TLP251 high-speed optocouplers driven via $56\ \Omega$ series resistors to safely isolate the low-voltage logic from the high-power MOSFETs.
- **Gate Drive Stage:** Bootstrap circuits using 63 V, 100 μ F capacitors to provide the floating gate voltage for the high-side MOSFETs.
- **Power Stage:** A standard H-bridge topology consisting of four IRF3205 N-channel MOSFETs.
- **Filter Stage:** A second-order LC low-pass filter ($L = 1\text{ mH}$, $C = 2.2\ \mu\text{F}$).

3.2 Design Calculations

3.2.1 Switching Frequency Selection

The switching frequency of 18 kHz was selected as a compromise between:

1. **Harmonic Performance:** Higher frequencies reduce filter size and improve THD.
2. **Switching Losses:** Higher frequencies increase switching losses in the IRF3205 MOSFETs.
3. **Microcontroller Capability:** The STM32F103 can comfortably generate PWM at 18 kHz.

3.2.2 Dead-Time Calculation

The minimum dead-time is calculated based on the TLP251 propagation delay and IRF3205 switching times:

$$t_{dead} = t_{d(off)} + t_f - t_{d(on)} \quad (3.1)$$

From the datasheets:

- TLP251 maximum propagation delay: 0.5 μs
- IRF3205 turn-off delay: 50 ns
- IRF3205 fall time: 40 ns

A conservative dead-time of 2.5 μs was implemented.

3.2.3 LC Filter Design

The LC filter cutoff frequency is:

$$f_c = \frac{1}{2\pi\sqrt{(1 \times 10^{-3} \text{ H})(2.2 \times 10^{-6} \text{ F})}} = 3.4 \text{ kHz} \quad (3.2)$$

The ratio of switching frequency to cutoff frequency is:

$$\frac{f_{sw}}{f_c} = \frac{18000}{3400} \approx 5.3 \quad (3.3)$$

Ideally, this ratio should be between 10 and 20 for good harmonic attenuation.

3.2.4 Bootstrap Capacitor Sizing

The bootstrap capacitor must maintain the high-side gate voltage during the on-time:

$$C_{boot} \gg \frac{Q_g}{\Delta V_{gs}} \quad (3.4)$$

For the IRF3205, $Q_g \approx 120 \text{ nC}$. With 100 μF , the voltage droop is minimal.

Chapter 4

Component Selection and Hardware Implementation

4.1 IRF3205 Power MOSFETs

According to the IRF3205 datasheet [1], this N-channel power MOSFET features an extremely low on-state resistance of $8\text{ m}\Omega$ typical at 10 V V_{GS} and a high continuous drain current capability of 110 A . These characteristics make it an excellent choice for low-voltage, high-current inverter applications. However, the datasheet also specifies a relatively high input capacitance ($C_{iss} \approx 3300\text{ pF}$), which necessitates a robust gate driver capable of supplying sufficient peak current to charge the gate capacitance rapidly.

4.1.1 Key Specifications

- $R_{DS(on)} = 8\text{ m}\Omega$ @ $V_{GS} = 10\text{ V}$
- $I_D = 110\text{ A}$ @ $T_C = 25^\circ\text{C}$
- $V_{GS} = \pm 20\text{ V}$ maximum
- $C_{iss} = 3300\text{ pF}$ typical
- $Q_g = 120\text{ nC}$ typical

4.2 TLP251 Optocoupler

The TLP251 is a high-speed optically isolated gate driver designed specifically for IGBTs and power MOSFETs. As per its datasheet [2], it provides a peak output current of up to 0.5 A and exhibits a typical propagation delay time of approximately $0.15\text{ }\mu\text{s}$.

4.2.1 Key Specifications

- Peak Output Current: $\pm 0.5\text{ A}$
- Propagation Delay: $0.15\text{ }\mu\text{s}$ typical
- Supply Voltage: 15 V to 30 V
- Input Current: 7 mA to 10 mA

4.3 STM32F103C8T6 Microcontroller

The STM32F103C8T6 is a 32-bit ARM Cortex-M3 microcontroller with:

- Clock speed: up to 72 MHz
- 64 KB Flash memory
- 20 KB SRAM
- Advanced timer modules for PWM generation
- 12-bit ADC for potential feedback implementation

4.4 PCB Design in EasyEDA

The schematic capture and PCB layout were executed in EasyEDA. Key considerations included:

1. **Power Routing:** Wide traces for high current paths.
2. **Grounding:** Star grounding to minimize noise.
3. **Decoupling:** Capacitors near power pins.
4. **Clearance:** Adequate spacing for high voltage isolation.

4.5 Complete Schematic

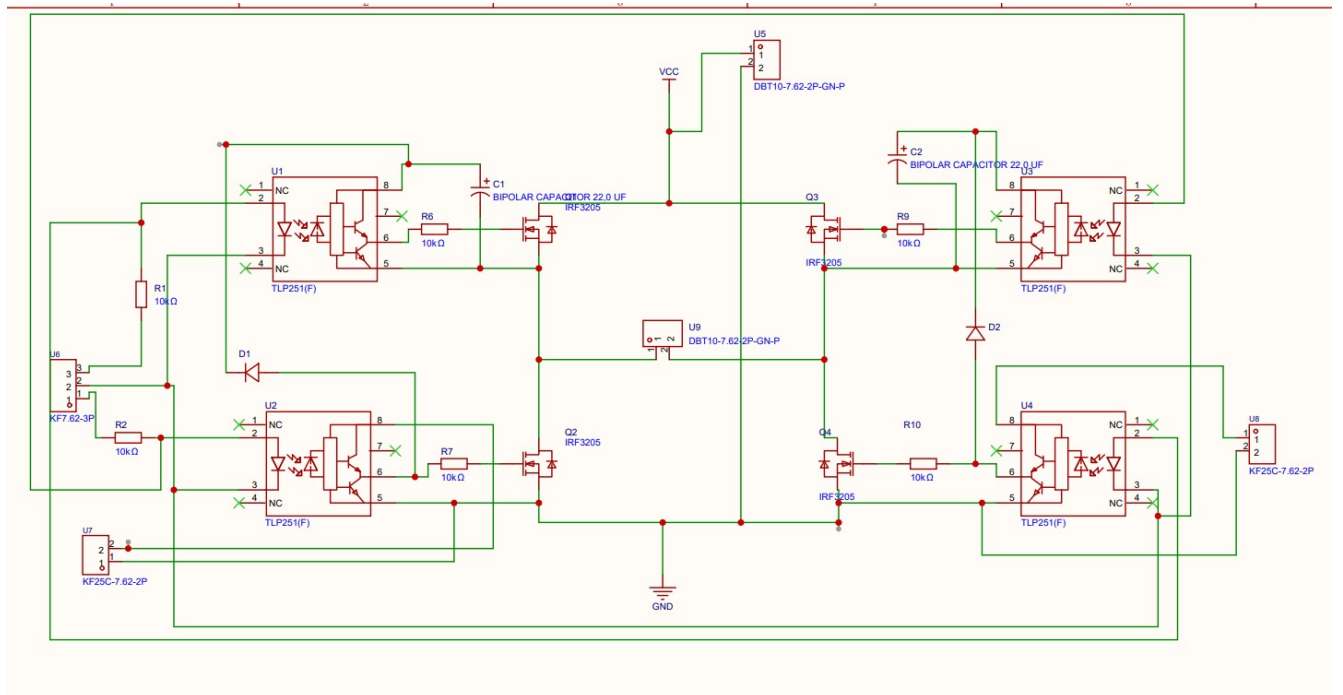


Figure 4.1: Complete Schematic Diagram of the SPWM Inverter

Chapter 5

Results and Observations

5.1 Experimental Setup

Testing was performed using a 4 V DC input. An oscilloscope was used to capture the output waveform under two distinct conditions:

- **Condition 1: No Load.** The output sine wave exhibited a clean, undistorted shape, confirming the correct generation of the SPWM algorithm and the basic functionality of the H-bridge.
- **Condition 2: Resistive Load ($2.2\ \Omega$).** Upon connecting the load, the waveform suffered distortion and a noticeable reduction in amplitude.

5.2 No-Load Results

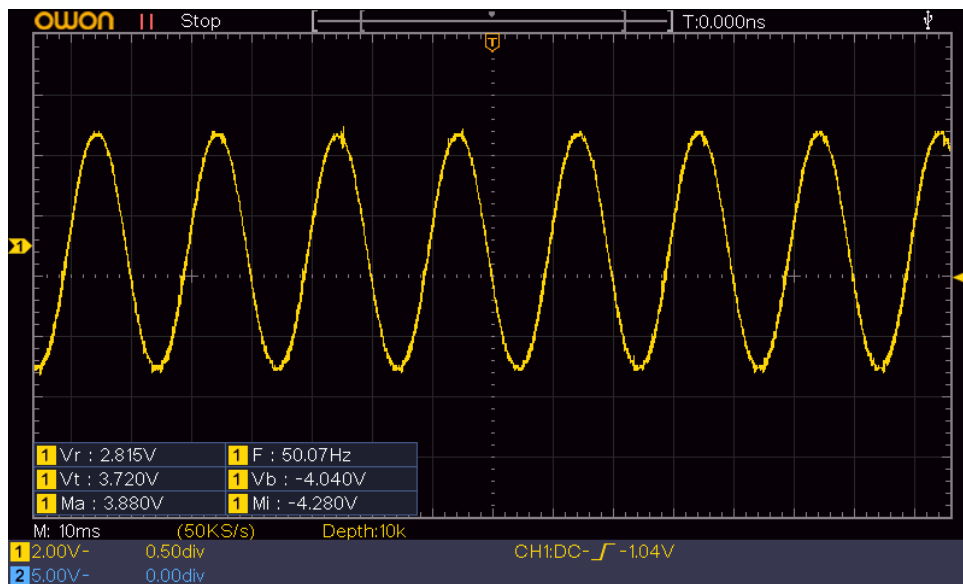


Figure 5.1: Output waveform under No-Load condition

Under no-load conditions:

- Peak output voltage: 3.7 V
- Waveform shape: Clean sinusoidal
- Frequency: 50 Hz

5.3 Load Results

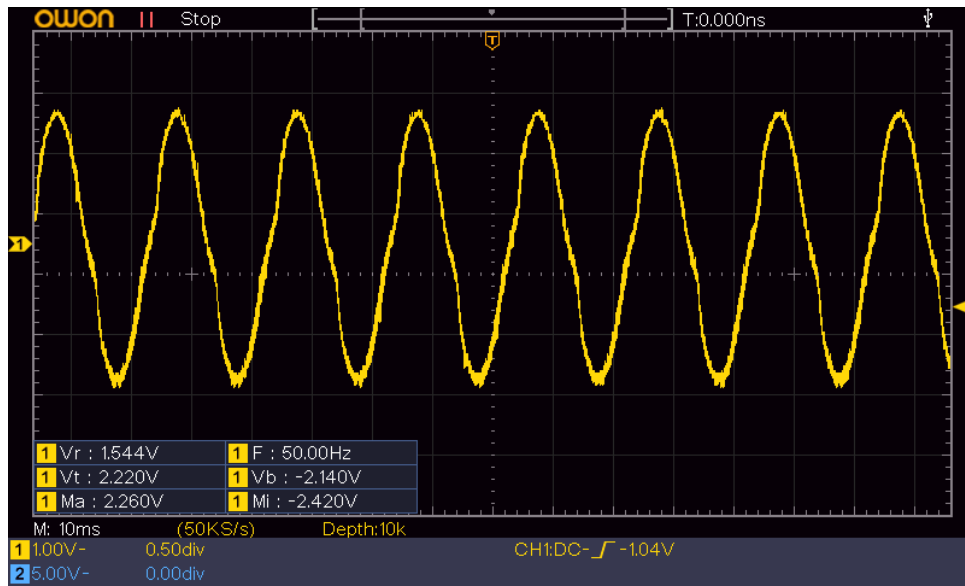


Figure 5.2: Output waveform with 2.2Ω resistive load attached

Under 2.2Ω load:

- Peak output voltage: 2.2 V
- Voltage sag: Approximately 40 %
- Waveform shape: Slight distortion observed

5.4 Hardware Setup

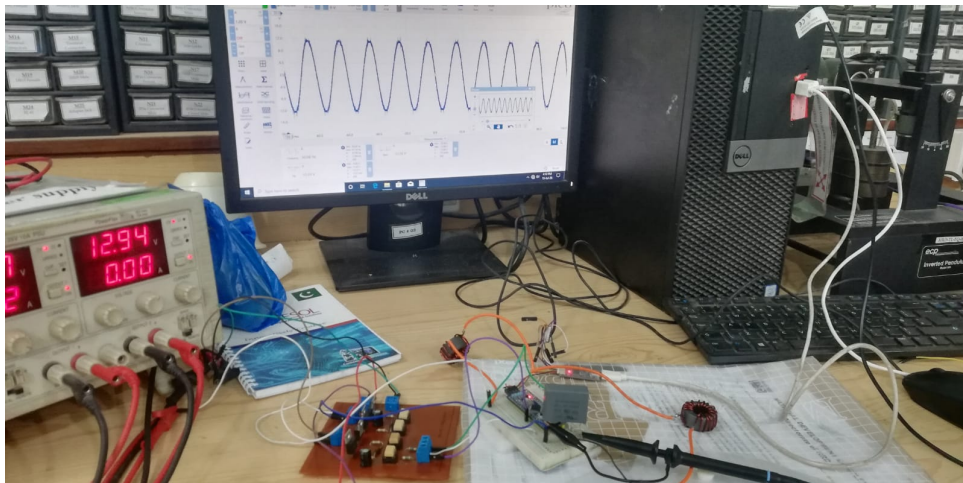


Figure 5.3: Physical test bench showing the inverter PCB, supplies, and oscilloscope

5.5 Summary of Results

Table 5.1: Summary of Experimental Results

Parameter	No Load	2.2 Load
Peak Output Voltage	3.7 V	2.2 V
Voltage Sag	–	40%
Waveform Quality	Clean	Slightly Distorted
Frequency	50 Hz	50 Hz

Chapter 6

Discussion

6.1 Analysis of Amplitude Sag

The most significant observation is the amplitude drop from 3.7 V peak under no-load to 2.2 V peak when a $2.2\ \Omega$ resistive load is connected. This represents a voltage sag of approximately 40 %.

The $56\ \Omega$ resistors used in this design are placed in series with the TLP251's input LED to limit the forward current from the STM32, as per the optocoupler's recommended operating conditions [2]. They have no effect on the MOSFET switching speed. The MOSFETs themselves are driven correctly at 12 V through the TLP251 and bootstrap circuit, allowing them to achieve their low $R_{DS(on)}$ of approximately $8\ \text{m}\Omega$.

6.1.1 Causes of Voltage Sag

The amplitude sag is primarily caused by two factors:

1. **Low DC Bus Voltage (4 V):** With a $2.2\ \Omega$ load, the current is approximately 1 A. Any resistance in the circuit becomes significant at such a low voltage. The main limitation is that the output cannot exceed the 4 V bus, and the load pulls the voltage down.
2. **DC Resistance (DCR) of the Inductor:** The 1 mH inductor has a winding resistance (DCR). Even a small DCR of $0.5\ \Omega$ to $1\ \Omega$ creates a substantial voltage drop when 1 A flows through it, acting as a voltage divider with the load.

6.2 Waveform Shape and Filter Response

Contrary to severe resonance issues, the waveform shape changed only slightly when the load was applied. This indicates that the LC filter ($1\ \text{mH} + 2.2\ \mu\text{F}$) is reasonably stable, but the high output impedance of the filter (due to the inductor's DCR) prevents the system from delivering sufficient current without dropping voltage. The slight shape change observed is likely due to the load altering the damping factor of the LC network.

6.3 Comparison with Theoretical Predictions

The theoretical cutoff frequency of the LC filter is 3.4 kHz. With a switching frequency of 18 kHz, the ratio $f_{sw}/f_c \approx 5.3$. Ideally, this ratio should be between 10 and 20 for good attenuation of switching harmonics. The filter's inadequate attenuation allows some high-frequency components to appear in the output, contributing to the slight waveform distortion.

6.4 Impact of Dead-Time on Performance

The implemented dead-time of $2.5\text{ }\mu\text{s}$ ensures safe operation and prevents shoot-through. However, as noted in the literature [5], dead-time introduces voltage error and low-order harmonics. At the low 4 V DC bus, the dead-time voltage error is more significant relative to the output voltage, contributing to the observed distortion.

6.5 Evaluation of Component Selection

6.5.1 IRF3205 Performance

The IRF3205 MOSFETs perform well, achieving low $R_{DS(on)}$ at 12 V gate drive. However, the low 4 V DC bus limits their effectiveness. The high input capacitance (3300 pF) requires adequate gate drive, which the TLP251 provides.

6.5.2 TLP251 Performance

The TLP251 optocouplers provide effective isolation and sufficient peak current for the IRF3205 gate capacitance. The propagation delay of $0.15\text{ }\mu\text{s}$ is well within acceptable limits.

6.5.3 LC Filter Performance

The 1 mH inductor with $2.2\text{ }\mu\text{F}$ capacitor provides basic filtering but is inadequate for full load conditions. The high DCR of the inductor significantly contributes to voltage sag.

Chapter 7

Design Challenges and Lessons Learned

7.1 Challenges Encountered

7.1.1 Filter Design

The primary challenge was designing a filter that provides adequate harmonic attenuation while maintaining reasonable output impedance. The 1 mH inductor, while available, had higher DCR than anticipated.

7.1.2 Load Regulation

Maintaining output voltage stability under varying loads proved challenging in open-loop configuration. The significant voltage sag under load indicates the need for closed-loop control.

7.1.3 Gate Drive Layout

PCB layout for the gate drive circuitry required careful attention to minimize parasitic inductance and avoid false triggering.

7.2 Lessons Learned

1. **Open-Loop Limitations:** Open-loop control is insufficient for applications requiring stable output under varying loads.
2. **Component Selection:** Careful attention to parasitic parameters (like inductor DCR) is essential.
3. **Design Margins:** Always provide adequate design margins for DC bus voltage and component ratings.
4. **Testing Methodology:** Systematic testing under multiple load conditions reveals design weaknesses.

Chapter 8

Future Work and Recommendations

8.1 Immediate Improvements

Based on the findings of this project, the following immediate improvements are recommended:

1. **Increase DC Bus Voltage:** Evaluate the system at 12 V or 24 V to fully utilize the low $R_{DS(on)}$ capability of the IRF3205.
2. **Improve LC Filter:** Replace the 1 mH inductor with a toroidal inductor having:
 - Higher inductance: 10 mH to 15 mH
 - Lower DCR: High current rating with thicker wire
3. **Implement Closed-Loop Control:** Integrate a voltage feedback network into the STM32 ADC and implement a digital PID controller.
4. **Add Damping Resistor:** Insert a $10\ \Omega$ to $22\ \Omega$ resistor in series with the $2.2\ \mu\text{F}$ capacitor.

8.2 Long-Term Development

For future iterations, the following enhancements are suggested:

1. **Dedicated Gate Driver IC:** Use a high-current gate driver (e.g., IR2110, UCC21222) for faster switching.
2. **Thermal Management:** Add heatsinks to the IRF3205 MOSFETs, especially when operating at higher power levels.
3. **THD Analysis:** Perform detailed THD analysis using an oscilloscope FFT function or dedicated analyzer.
4. **Digital Control:** Implement more advanced control algorithms like Repetitive Control or Predictive Control.
5. **Protection Features:** Add over-current, over-voltage, and over-temperature protection.

Chapter 9

Conclusion

This internship was a valuable hands-on experience that covered the full process of building an embedded power electronics project. The work included writing the control algorithm, designing the schematic and PCB in EasyEDA, assembling the components, and testing the hardware thoroughly. The final prototype successfully produces SPWM signals to drive an H-bridge, proving that the microcontroller-based design works.

The load testing showed a major issue with voltage regulation. The output voltage dropped from 3.7 V peak with no load to 2.2 V peak when a 2.2Ω load was connected. At low voltage, even a small resistance in the inductor or the MOSFETs causes a noticeable voltage loss when the load draws current. The waveform shape changed only slightly, indicating that the LC filter is reasonably stable but has high output impedance.

To build a stable and well-regulated inverter in the future, the main DC bus voltage should be increased to a higher value, such as 12 V or 24 V. A closed-loop feedback system with a PID controller should also be added to automatically adjust the output and compensate for any voltage drop under load. The LC filter should be improved using an inductor with lower DC resistance as well.

The objectives of the internship were successfully achieved:

1. A complete SPWM inverter system was designed and implemented.
2. Complementary SPWM signals with proper dead-time were generated and verified.
3. An LC filter was designed and its performance evaluated.
4. Load testing identified key limitations and provided recommendations for improvement.

These lessons learned meet the project's learning goals and provide a good starting point for future work toward a commercial-quality inverter. The experience gained in PCB design, microcontroller programming, power electronics, and hardware debugging will be invaluable for future engineering projects.

Acknowledgment

I extend my sincere gratitude to Dr. Raheel Qureshi for their continuous guidance and support throughout this internship. Their expertise in power electronics was instrumental in navigating the design challenges encountered. I also thank the engineering team at NESCOM for providing the necessary resources, lab facilities, and technical support to conduct this project. The collaborative environment and access to professional equipment significantly enhanced the quality and scope of this work.

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Appendix

Appendix A: Component Datasheets

The detailed manufacturer datasheets for the following components are attached for reference:

- IRF3205 N-Channel Power MOSFET (International Rectifier)
- TLP251 High-Speed Optocoupler (Toshiba)

These documents contain the absolute maximum ratings, electrical characteristics, switching characteristics, and thermal data that justify the component selection and calculations presented in this report.

IRF3205 Datasheet Specifications

Table 9.1: Absolute Maximum Ratings (IRF3205) [1]

Parameter	Value	Unit
Continuous Drain Current, $V_{GS} = 10V$ @ $T_C = 25^\circ C$	110	A
Continuous Drain Current, $V_{GS} = 10V$ @ $T_C = 100^\circ C$	80	A
Pulsed Drain Current, I_{DM}	390	A
Power Dissipation @ $T_C = 25^\circ C$	200	W
Linear Derating Factor	1.3	W/ $^\circ C$
Gate-to-Source Voltage, V_{GS}	± 20	V
Avalanche Current, I_{AR}	62	A
Repetitive Avalanche Energy, E_{AR}	20	mJ
Peak Diode Recovery dv/dt	5.0	V/ns
Operating Junction & Storage Temperature Range, T_J, T_{STG}	-55 to +175	$^\circ C$
Soldering Temperature (for 10 seconds, 1.6mm from case)	300	$^\circ C$
Mounting Torque (6-32 or M3 screw)	10 lbf·in (1.1 N·m)	–

Table 9.2: Thermal Resistance (IRF3205) [1]

Parameter	Symbol	Typ.	Max.
Junction-to-Case	$R_{\theta JC}$	–	0.75
Case-to-Sink (Flat, Greased Surface)	$R_{\theta CS}$	0.50	–
Junction-to-Ambient	$R_{\theta JA}$	–	62

TLP251 Datasheet Specifications

Table 9.3: Recommended Operating Conditions (TLP251) [2]

Characteristic	Symbol	Min.	Typ.	Max.
Input Current (ON)	$I_{F(ON)}$	7	8	10
Input Voltage (OFF)	$V_{F(OFF)}$	0	–	0.8
Supply Voltage	V_{CC}	15	–	30
Peak Output Current	I_{OPH}/I_{OPL}	–	–	± 0.5
Operating Temperature	T_{OPR}	-20	25	85

Table 9.4: Electrical Characteristics (TLP251) ($T_a = -20$ to 70°C , unless otherwise specified) [2]

Characteristic	Symbol	Test Condition	Min.	Typ.*	Max.
Input Forward Voltage	V_F	$I_F = 10\text{ mA}$, $T_a = 25^\circ\text{C}$	–	1.6	1.8
Temp. Coefficient of Forward Voltage	$\Delta V_F / \Delta T_a$	$I_F = 10\text{ mA}$	–	-2.0	–
Input Reverse Current	I_R	$V_R = 5\text{ V}$, $T_a = 25^\circ\text{C}$	–	–	10
Input Capacitance	C_T	$V = 0\text{ V}$, $f = 1\text{ MHz}$, $T_a = 25^\circ\text{C}$	–	45	250
Output Current (H level)	I_{OPH}	$I_F = 10\text{ mA}$, $V_{8-6} = 4\text{ V}$	-0.5	-1.5	–
Output Current (L level)	I_{OPL}	$I_F = 0\text{ mA}$, $V_{6-5} = 2.5\text{ V}$	0.5	2	–
Output Voltage (H level)	V_{OH}	$V_{CC1} = +15\text{ V}$, $V_{EE1} = -15\text{ V}$, $R_L = 200\Omega$, $I_F = 5\text{ mA}$	11	12.8	–
Output Voltage (L level)	V_{OL}	$V_{CC1} = +15\text{ V}$, $V_{EE1} = -15\text{ V}$, $R_L = 200\Omega$, $V_F = 0.8\text{ V}$	–	-14.2	-12.5

* All typical values are at $T_a = 25^\circ\text{C}$.

Table 9.5: Switching Characteristics (TLP251) ($T_a = -20$ to 70°C , unless otherwise specified) [2]

Characteristic	Symbol	Test Min. Con- di- tion	Typ.	Max.	Unit
Propagation Delay (L $\rightarrow$ H)	t_{pLH}	$I_F = -$ 8 mA, $V_{CC} =$ 15 V, $R_L =$ 200 Ω	0.15	0.5	μs
Propagation Delay (H $\rightarrow$ L)	t_{pHL}	$I_F = -$ 8 mA, $V_{CC} =$ 15 V, $R_L =$ 200 Ω	0.15	0.5	μs
Common Mode Transient Immunity (H level)	CM_H	$V_{CM} = 5000$ 600 V, $I_F =$ 8 mA, $V_{CC} =$ 30 V, $T_a =$ 25 $^\circ\text{C}$	—	—	V/ μs
Common Mode Transient Immunity (L level)	CM_L	$V_{CM} = 5000$ 600 V, $I_F =$ 0 mA, $V_{CC} =$ 30 V, $T_a =$ 25 $^\circ\text{C}$	—	—	V/ μs

* All typical values are at $T_a = 25^\circ\text{C}$.